

Unit II Neural Networks and Training

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1. Neural Network Kya Hai?

Neural network biological brain se inspire hokar banaya gaya computational model hai. Isme neurons ke groups form karte hain aur inke through information process hota hai.

Basic Structure

? Input layer

? Hidden layers

? Output layer

Main Idea

Input data ko process karke output generate karna.

2. Perceptron

Perceptron neural network ka simplest form hai. Ye input values ko weights aur bias ke saath combine karta hai aur output generate karta hai.

Formula

Output roughly kisi threshold function ke basis par decide hota hai.

3. Weights and Bias

Weights

Weights inputs ko adjust karte hain. Ye model ki learning ko control karte hain.

Bias

Bias neuron ko shift karne mein help karta hai. Isse model ko better fit dene mein madad milti hai.

4. Activation Functions

Activation functions non-linearity introduce karte hain. Ye neural network ko complex patterns samajhne mein help karte hain.

Common Activation Functions

? Sigmoid

? Tanh

? ReLU

? Softmax

Why Important

Without activation functions, neural network sirf linear relationships hi samajh sakta hai.

5. Loss Function

Loss function model ke predicted output aur actual output ke beech error measure karta hai.

Common Loss Functions

? Mean Squared Error

? Cross-Entropy Loss

Purpose

Loss kam karna hi training ka goal hai.

6. Gradient Descent

Gradient descent ek optimization technique hai jisme weights ko update karke loss ko minimize kiya jata hai.

Main Idea

Loss function ke gradient ka use karna aur parameters ko us direction mein adjust karna jahan loss kam ho.

Types

- ? Batch Gradient Descent
- ? Stochastic Gradient Descent
- ? Mini-batch Gradient Descent

7. Backpropagation

Backpropagation training ka core process hai. Isme error ko output layer se input layer tak propagate kiya jata hai aur weights update hote hain.

Importance

Ye deep networks ko train karne ke liye bohot important hai.

8. Multilayer Neural Network

Jab neural network mein multiple hidden layers hote hain, tab use deep neural network kehte hain. Ye complex patterns samajh sakta hai.

Uses

- ? Image recognition
- ? NLP
- ? Speech processing

9. Weight Initialization

Initial weights ka selection training ko directly affect karta hai. Agar weights bohot bade ya bohot chhote hon to training weak ho sakti hai.

Common Methods

- ? Xavier initialization
- ? He initialization

10. Training, Validation and Testing

Training

Model ko data par learn karaya jata hai.

Validation

Model ko tune karne ke liye validation data use hota hai.

Testing

Model ko unseen data par test kiya jata hai.

11. Vanishing and Exploding Gradient Problem

Agar gradients bahut chhote ya bahut bade ho jaye to training unstable ho sakti hai.

Solutions

? ReLU activation

? Proper initialization

? Batch normalization

12. Regularization Techniques

Regularization overfitting ko reduce karne ke liye use hoti hai.

L1 Regularization

? Sparsity laata hai

? Feature selection mein useful

L2 Regularization

? Weights ko small rakhta hai

? Model ko simple banata hai

Dropout

- ? Randomly neurons ko off karta hai
- ? Overfitting ko kam karta hai

Batch Normalization

- ? Training ko stable banata hai

13. Autoencoders

Autoencoder input ko compress karke phir reconstruct karta hai. Ye dimensionality reduction aur feature learning mein use hota hai.

Main Parts

- ? Encoder
- ? Decoder

14. Hyperparameter Tuning

Hyperparameters jaise learning rate, batch size, epochs aur number of layers ka tuning model performance ko improve karta hai.

Important Hyperparameters

- ? Learning rate
- ? Batch size
- ? Epochs
- ? Number of layers

15. Important Exam Points

- ? Neural network mein input, hidden aur output layers hote hain.
- ? Weights aur bias model ki learning ko define karte hain.
- ? Activation functions non-linearity introduce karte hain.
- ? Gradient descent aur backpropagation training ke core concepts hain.
- ? Regularization overfitting ko reduce karta hai.